Solid state physics – crystals, lattices and bands

Why does copper conduct electricity while diamond does not? Why is silicon a semiconductor, and why is diamond transparent and copper reddish? Solid state physics explains the properties of materials from two simple ideas: the atoms sit in a regular lattice, and the electrons in such a lattice behave as waves that can only have certain energies – bands. These ideas made the transistor, the solar cell, the LED and the computer possible. This page covers the foundations of solid state physics with three interactive figures.

Crystals and lattices

As early as 1669 the Danish naturalist Niels Stensen (Nicolaus Steno) noticed that the angles between the faces of rock crystal are always the same, whatever the size and shape of the crystal. Around 1784 René Just Haüy suggested that crystals are built of identical small building blocks repeated in all directions. In 1848 Auguste Bravais showed that in three dimensions there are only 14 different ways of arranging points regularly – the 14 Bravais lattices.

A crystal is described by a lattice and a *basis*: the group of atoms that sits at each lattice point. Three cubic lattices are especially important:

Simple cubic (sc)

One atom at each corner of a cube. Rare – only polonium crystallises like this.

Body-centred cubic (bcc)

An extra atom in the centre of the cube. Iron at room temperature, sodium, tungsten.

Face-centred cubic (fcc)

An extra atom in the middle of each face – the densest packing of spheres. Copper, aluminium, gold, silver. Diamond and silicon are fcc with a basis of two atoms.

Miller indices

Planes in the crystal are labelled by three integers (hkl). The spacing of the planes in a cubic lattice with lattice constant a is $d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$.

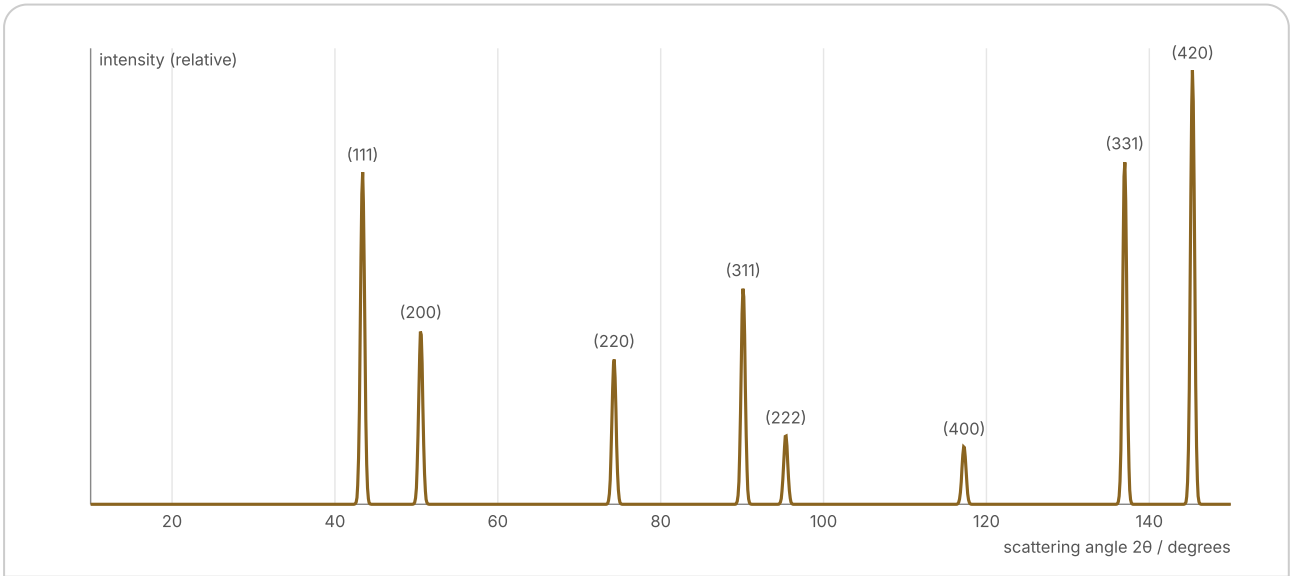
X-ray diffraction

In 1912 Max von Laue suggested that crystals might act as a three-dimensional grating for the newly discovered X-rays, since the spacing between atoms is of the same size as the wavelength. Walter Friedrich and Paul Knipping tried it with a crystal of copper sulphate – and got a pattern of spots. This proved at once that X-rays are waves and that crystals are periodic. The same year the 22-year-old Lawrence Bragg in Cambridge formulated a simple rule: the X-rays are reflected from the atomic planes, and the reflections reinforce each other when the path difference is a whole number of wavelengths:

$$2d \sin\theta = n\lambda$$

Together with his father, William Henry Bragg, he determined the first crystal structures – rock salt, diamond, copper. Father and son shared the Nobel Prize in 1915; Lawrence is still the youngest Nobel laureate in physics. X-ray crystallography was later used to determine the structure of penicillin, DNA (Rosalind Franklin's Photo 51, 1952) and haemoglobin.

Not all planes give a reflection. In a bcc lattice the contributions from the corner and centre atoms cancel unless $h + k + l$ is even. In fcc, h, k, l must be either all even or all odd. The pattern of missing lines therefore reveals the type of lattice.



The reciprocal lattice

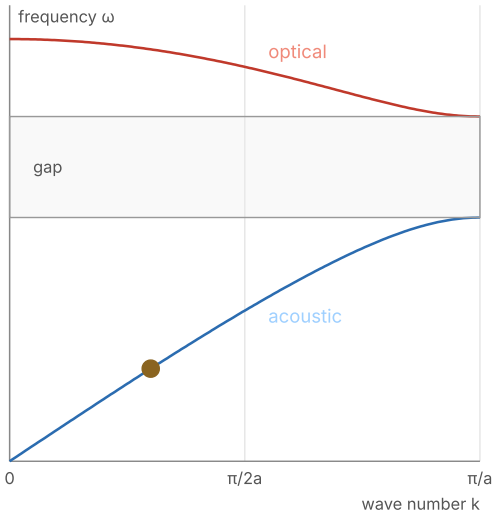
To calculate diffraction in an arbitrary crystal it is convenient to introduce the reciprocal lattice with basis vectors $\vec{b}_i$, defined by $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$. Each point $\vec{G}$ of the reciprocal lattice corresponds to a family of lattice planes, and the Laue condition states that there is a reflection when the change in wave vector is a reciprocal lattice vector: $\vec{k}\,' - \vec{k} = \vec{G}$. The reciprocal lattice is the natural "space" for all waves in the crystal – X-rays, lattice vibrations and electrons. The region around the origin that is closer to the origin than to any other lattice point is called the first *Brillouin zone*.

Lattice vibrations and phonons

The atoms in a crystal do not sit still; they vibrate about their equilibrium positions, connected by forces like springs. Waves running through the lattice are sound at long wavelengths, but at short wavelengths they feel that the lattice is discrete. For a chain with two different atoms in each cell (masses m_1 and m_2, spring constant C) there are two branches:

$$\omega^2 = C\left(\frac{1}{m_1}+\frac{1}{m_2}\right) \pm C\sqrt{\left(\frac{1}{m_1}+\frac{1}{m_2}\right)^2 - \frac{4\sin^2(ka/2)}{m_1 m_2}}$$

The *acoustic* branch corresponds to neighbouring atoms moving in step (sound), the *optical* branch to their moving in opposition – in ionic crystals such as NaCl it can be excited by infrared light. Between the branches there is a frequency gap. The vibrations are quantised; the quanta are called *phonons*.





atom 1 (gold) and atom 2 (blue)

Heat capacity: Einstein and Debye

In 1819 Dulong and Petit found that the heat capacity of solids is about $3R \approx 25$ J/(mol·K) – as classical equipartition predicts. But at low temperatures it falls towards zero, and diamond has an unusually low heat capacity already at room temperature. In 1907 Einstein applied Planck's quantum hypothesis to the lattice vibrations: if all atoms vibrate at the same frequency, the vibrations "freeze out" at low temperatures. It was one of the first applications of quantum theory outside radiation. In 1912 Peter Debye improved the model by including the whole spectrum of sound waves and predicted that the heat capacity goes as T^3 at low temperatures – in fine agreement with measurement. See also Statistical mechanics.

Free electrons

In 1900 Paul Drude proposed that the conductivity of metals is due to a gas of free electrons colliding with the ions. The model explained Ohm's law and the Wiedemann–Franz law, but not why the electrons contribute almost nothing to the heat capacity. The answer came in 1927, when Arnold Sommerfeld applied Fermi–Dirac statistics: because of the exclusion principle the electrons are filled up to a high energy, the Fermi energy E_F, typically a few eV – equivalent to tens of thousands of degrees. At room temperature only the electrons within about $k_B T$ of the Fermi level can change state.

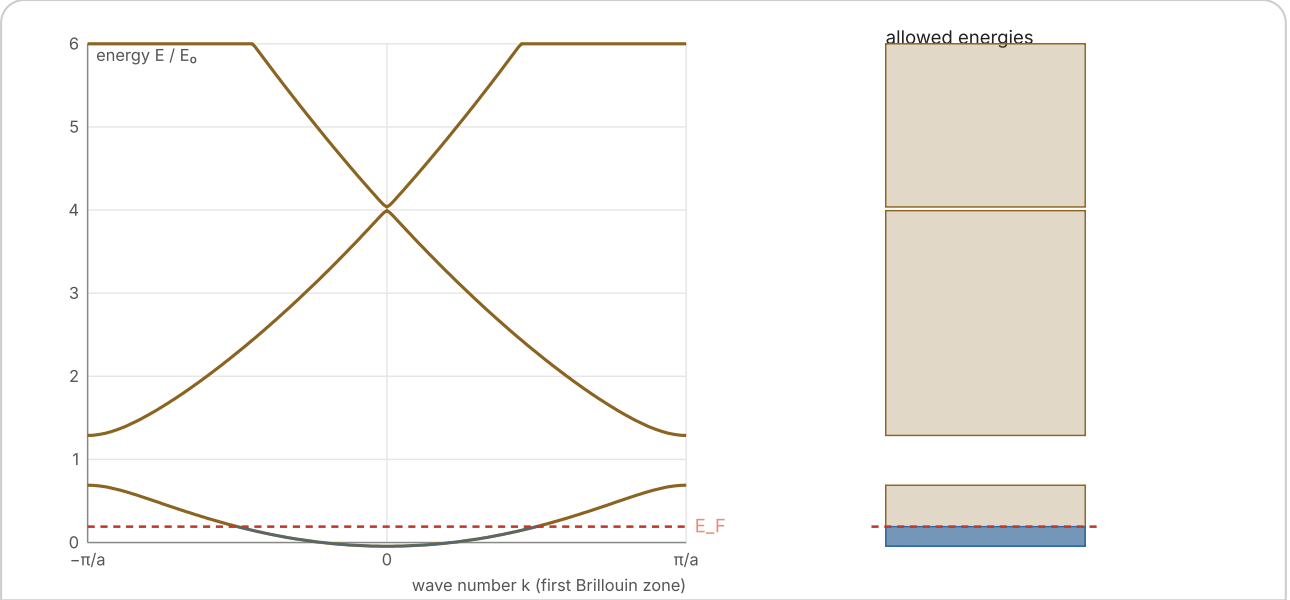
$$E_F = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}, \qquad \sigma = \frac{n e^2 \tau}{m}$$

Bands: why some materials conduct and others do not

In 1928 the 23-year-old Felix Bloch showed in his doctoral thesis under Heisenberg that an electron in a periodic potential can be described as a plane wave multiplied by a function with the period of the lattice. The electron moves through a perfect lattice without being scattered – resistance is due only to impurities and lattice vibrations.

$$\psi_{\vec{k}}(\vec{r}) = e^{i\vec{k}\cdot\vec{r}}\, u_{\vec{k}}(\vec{r}), \qquad u_{\vec{k}}(\vec{r}+\vec{R}) = u_{\vec{k}}(\vec{r})$$

At the edge of the Brillouin zone the electron wave satisfies Bragg's condition and is reflected. Two standing waves with different energies arise, and between them an energy gap. The allowed energies collect in *bands*. In 1931 Alan Wilson explained the consequence: if the highest band containing electrons is only partly filled, the electrons can easily move into empty states, and the material is a metal. If it is completely filled and the next band lies a gap above, the material is an insulator – or a semiconductor if the gap is small.



Semiconductors and the transistor

In a semiconductor such as silicon the band gap is 1.12 eV. At room temperature a small number of electrons are excited across the gap; they leave behind *holes* in the valence band, which behave as positive charges. The conductivity can be changed by many orders of magnitude by adding small amounts of impurities – *doping*. Phosphorus has one more valence electron than silicon and gives extra electrons (n-type); boron has one fewer and gives holes (p-type).

Where a p-region meets an n-region, a diode forms that conducts current only one way. On 23 December 1947 John Bardeen and Walter Brattain at Bell Labs demonstrated the first transistor, a point-contact transistor of germanium; William Shockley invented the junction transistor shortly afterwards. The three shared the Nobel Prize in 1956. Jack Kilby and Robert Noyce made the first integrated circuits in 1958–59. Today a processor contains many billions of transistors.

Semiconductor physics also gives us light-emitting diodes: when an electron in the conduction band drops into a hole, the energy can be emitted as a photon with an energy equal to the band gap. Blue LEDs of gallium nitride, developed by Akasaki, Amano and Nakamura (Nobel Prize 2014), made white LED lamps possible. Solar cells are LEDs in reverse. See Laser and Laser physics.

Magnetism

Their spin makes electrons tiny magnets. In most materials they point at random, but in iron, cobalt and nickel a quantum-mechanical effect – the *exchange interaction*, explained by Heisenberg in 1928 – makes neighbouring spins line up in parallel. Below the Curie temperature (770 °C for iron) the material divides into magnetic domains. Pierre Weiss's mean-field theory of 1907 and the Ising model (see Statistical mechanics) are the classical models. Louis Néel predicted antiferromagnetism, in which neighbouring spins point in opposite directions. Giant

magnetoresistance, discovered by Albert Fert and Peter Grünberg in 1988 (Nobel Prize 2007), made large hard disks possible.

New materials

Superconductors

Below a critical temperature the electrons form pairs and conduct without resistance. See Superconductivity and superfluidity.

Graphene

A single layer of carbon atoms in a hexagonal lattice, isolated with sticky tape by Andre Geim and Konstantin Novoselov in 2004 (Nobel Prize 2010). The electrons behave like massless particles.

Quasicrystals

In 1982 Dan Shechtman saw a diffraction pattern with fivefold symmetry, which is impossible in a periodic lattice. He was ridiculed for years but received the Nobel Prize in 2011.

Topological insulators

Materials that insulate inside but conduct on the surface, protected by topology. Thouless, Haldane and Kosterlitz received the Nobel Prize in 2016 for the theory.

Further reading

- Charles Kittel: *Introduction to Solid State Physics* (8th ed., Wiley 2005)
- Neil W. Ashcroft and N. David Mermin: *Solid State Physics* (1976)
- H.M. Rosenberg: *The Solid State* (3rd ed., Oxford 1988)
- Steven H. Simon: *The Oxford Solid State Basics* (Oxford 2013)

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